

Conducting Multiple Linear Regression in R

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Hochschule Fresenius

Course: Data Analysis for Decision-Making

Professor: Prof. Dr. Stephan Huber

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1 Scope of Presentation

This presentation covers:

- How to **load and inspect** data in R
- How to **specify and run** a multiple linear regression model using `lm()`
- How to **extract and interpret** results from `summary()`
- How to run basic **model diagnostics**

i Note

Theory of regression is assumed as background knowledge.

2 Workflow Overview

1. Data Input

Load & inspect

2. Model Specification

Define formula

3. Model Estimation

Fit with `lm()`

4. Output Extraction

`summary()` & plots

Each step builds on the previous one — a clear, repeatable workflow.

3 Data Input in R

Create a data frame directly in R:

```
data <- data.frame(  
  Salary    = c(2000, 2500, 3000, 3500, 4000),  
  Age       = c(22, 25, 30, 35, 40),  
  Education = c(12, 14, 16, 16, 18),  
  Experience = c(1, 3, 5, 7, 10)  
)
```

Tip

In practice, import data from CSV with `read.csv("file.csv")` or from Excel with `readxl::read_excel("file.xlsx")`.

4 Data Inspection

Use these built-in functions before modeling:

```
head(data)      # shows the first 6 rows; confirms data loaded correctly
```

	Salary	Age	Education	Experience
1	2000	22	12	1
2	2500	25	14	3
3	3000	30	16	5
4	3500	35	16	7
5	4000	40	18	10

```
summary(data)    # gives min, max, mean, and quartiles for each variable
```

Salary		Age		Education		Experience	
Min.	:2000	Min.	:22.0	Min.	:12.0	Min.	: 1.0
1st Qu.	:2500	1st Qu.	:25.0	1st Qu.	:14.0	1st Qu.	: 3.0
Median	:3000	Median	:30.0	Median	:16.0	Median	: 5.0
Mean	:3000	Mean	:30.4	Mean	:15.2	Mean	: 5.2
3rd Qu.	:3500	3rd Qu.	:35.0	3rd Qu.	:16.0	3rd Qu.	: 7.0
Max.	:4000	Max.	:40.0	Max.	:18.0	Max.	:10.0

```
str(data)          # checks variable types (numeric, factor, etc.)
```

```
'data.frame':  5 obs. of  4 variables:
 $ Salary      : num  2000 2500 3000 3500 4000
 $ Age         : num   22  25  30  35  40
 $ Education   : num   12  14  16  16  18
 $ Experience  : num    1  3  5  7  10
```

```
is.na(data)        # check for missing values
```

```
      Salary  Age Education Experience
[1,] FALSE FALSE      FALSE      FALSE
[2,] FALSE FALSE      FALSE      FALSE
[3,] FALSE FALSE      FALSE      FALSE
[4,] FALSE FALSE      FALSE      FALSE
[5,] FALSE FALSE      FALSE      FALSE
```

Example: `summary(data)` reveals average salary is 3000, education ranges 12–18 years, experience 1–10 years — useful context before modeling.

5 Model Specification

Formula: `Salary ~ Age + Education + Experience`

Symbol	Meaning
~	Separates outcome from predictors
Left side	Dependent variable (y): Salary
Right side	Independent variables (x): Age, Education, Experience

```
# The formula interface in R
Salary ~ Age + Education + Experience
```

```
Salary ~ Age + Education + Experience
```

6 Model Estimation

Fit the model using **Ordinary Least Squares (OLS)**:

```
model <- lm(Salary ~ Age + Education + Experience, data = data)
```

The `lm()` call returns a **model object** containing:

- Estimated coefficients
- Residuals and fitted values
- Diagnostic information

```
fitted(model)      # predicted values
```

1	2	3	4	5
2040.984	2450.820	3024.590	3450.820	4032.787

```
residuals(model)   # observed minus predicted
```

1	2	3	4	5
-40.98361	49.18033	-24.59016	49.18033	-32.78689

```
summary(model)     # full results table
```

Call:

```
lm(formula = Salary ~ Age + Education + Experience, data = data)
```

Residuals:

1	2	3	4	5
-40.98	49.18	-24.59	49.18	-32.79

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-655.738	2029.578	-0.323	0.801
Age	81.967	74.676	1.098	0.470
Education	73.770	82.783	0.891	0.537
Experience	8.197	180.886	0.045	0.971

Residual standard error: 90.54 on 1 degrees of freedom

Multiple R-squared: 0.9967, Adjusted R-squared: 0.9869

F-statistic: 101.3 on 3 and 1 DF, p-value: 0.07287

7 Extracting Results

```
summary(model)
```

Call:

```
lm(formula = Salary ~ Age + Education + Experience, data = data)
```

Residuals:

```
      1      2      3      4      5  
-40.98  49.18 -24.59  49.18 -32.79
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)  
(Intercept) -655.738    2029.578  -0.323   0.801  
Age           81.967     74.676   1.098   0.470  
Education     73.770     82.783   0.891   0.537  
Experience     8.197    180.886   0.045   0.971
```

Residual standard error: 90.54 on 1 degrees of freedom

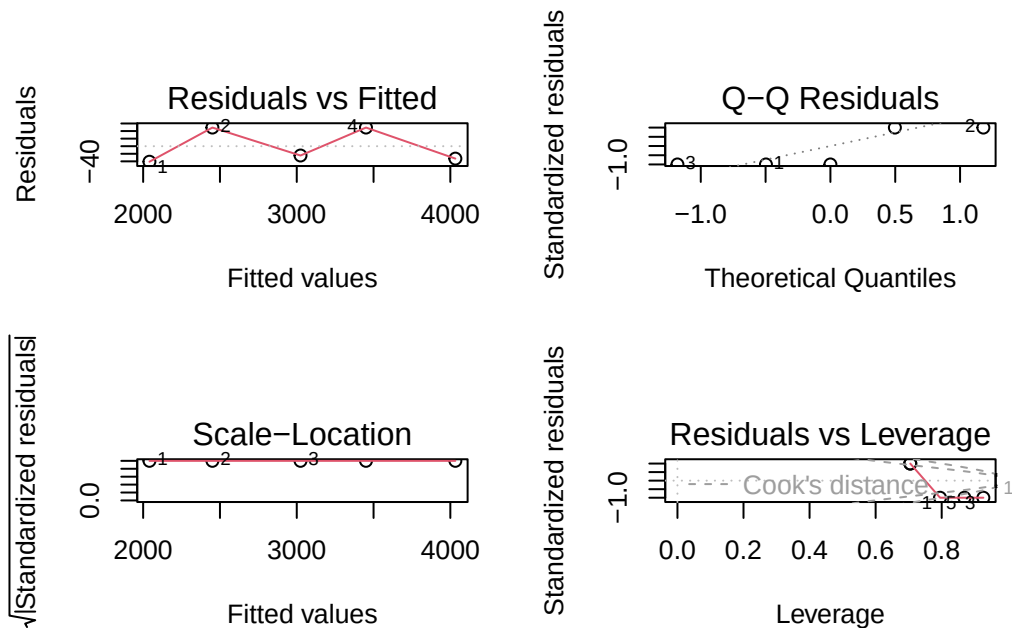
Multiple R-squared: 0.9967, Adjusted R-squared: 0.9869

F-statistic: 101.3 on 3 and 1 DF, p-value: 0.07287

Component	Interpretation
Estimate	Expected change in Y per one-unit increase in X, holding others constant
Std. Error	Precision of the coefficient estimate
t value	Estimate ÷ Std. Error; larger = stronger evidence
Pr(> t)	p-value; $p < 0.05 \rightarrow$ statistically significant
R-squared	Proportion of variance in Y explained by the model
Adjusted R²	R ² penalized for number of predictors
F-statistic	Tests if at least one predictor has a non-zero coefficient

8 Basic Diagnostics

```
par(mfrow = c(2, 2)) # display 4 plots in a 2x2 grid
plot(model)
```



What `plot(model)` shows:

- **Residuals vs Fitted** — checks linearity
- **Normal Q-Q** — checks if residuals are normally distributed
- **Scale-Location** — checks constant variance (homoscedasticity)
- **Cook's Distance** — detects influential outliers

9 Implementation Example

Predict salary for a **new employee**:

```
new_employee <- data.frame(
  Age      = 33,
  Education = 16,
  Experience = 6
)

# Point estimate
predict(model, newdata = new_employee)
```

```
1
3278.689
```

```
# With 95% prediction interval
predict(model, newdata = new_employee,
        interval = "prediction", level = 0.95)
```

```
      fit      lwr      upr
1 3278.689 1641.855 4915.522
```

i Note

The prediction interval gives both a point estimate and a plausible range — useful for decision-making.

10 Class Exercise

Run a regression model in R using `lm()` and interpret the output.

Use your own variables and the following structure:

```
# 1. Create your dataset (at least 8 observations, 2+ predictors)
my_data <- data.frame(
  y = c(...), # your outcome variable
  x1 = c(...), # first predictor
  x2 = c(...)  # second predictor
)
```



```
# 2. Fit the model
my_model <- lm(y ~ x1 + x2, data = my_data)

# 3. Inspect results
summary(my_model)

# 4. Diagnostics
par(mfrow = c(2, 2)); plot(my_model)

# 5. Predict a new observation
predict(my_model, newdata = data.frame(x1 = ..., x2 = ...))
```

Questions to answer: Which predictors are significant? What does adjusted R^2 tell you? Interpret one coefficient in plain language.

11 Key Takeaway

Multiple linear regression in R follows a **clear, repeatable workflow**:

Load data → **Inspect** → **Specify model with `lm()`**
 → **Extract results with `summary()`** → **Check diagnostics with `plot()`**

Mastering this sequence allows you to efficiently analyze relationships between variables and interpret **coefficients**, **p-values**, and **R-squared** in your own research.

12 References

- R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
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13 Thank You

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Questions welcome!